

Daniel DeAlcala

PhD candidate in AI, Universidad Autónoma de Madrid · Visiting Researcher, IBM Research – Zurich

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I work on vision-language models for document intelligence: turning pages into structured text and code. I am currently a visiting researcher with the Docling team at IBM Research Zurich, contributing to the open-source project; in 2025, at IBM Research in New York, I built a benchmark and a VLM detector for AI-edited documents. My PhD thesis is on auditing AI models, where I introduced the Membership Inference Test (MINT). 16 peer-reviewed papers, 9 as first author, including ICCV 2025 and IEEE Access.

RESEARCH EXPERIENCE

IBM Research – Zurich, AI for Knowledge

Zurich, Switzerland

Visiting Researcher, Docling team, hosted by Peter Staar

Sep 2026 – Dec 2026

- Contributor to Docling, IBM's open-source document conversion toolkit (60K+ GitHub stars), and working on several research projects around it.
- Developing a VLM that links the fields of a form to their values.

IBM Research, MIT-IBM Watson AI Lab

New York, USA

Visiting Researcher, hosted by Pin-Yu Chen

Sep 2025 – Dec 2025

- Took AI-generated content detection from single images or text passages to whole documents. Built **AidocFactory**, an open-source Python pipeline that parses PDFs with Docling, rewrites chosen paragraphs, figures and tables with any Hugging Face generative model, re-renders the page and records box-level labels. Used it to build a 35K-page benchmark from 2,000 arXiv papers.
- Trained **AidocDetector**, a 256M-parameter VLM that classifies each page and segments the edited regions. F1 of 0.83 on both seen and unseen generators and Dice of 0.85–0.88, above GPT-5.2, Gemini 3 Flash and Claude Opus 4.5 prompted as detectors.
- First-author paper in submission, with a journal extension in preparation; IBM filed a patent application on the method.

BiometricsAI Lab (formerly BiDA Lab), Universidad Autónoma de Madrid

Madrid, Spain

PhD Researcher and FPU Fellow, advised by Julian Fierrez and Aythami Morales

Jan 2022 – present

- Designed and trained the UAM team's entry to the DeepID Challenge at ICCV 2025 on forged ID documents: TruFor fine-tuned with a SegFormer backbone and knowledge distillation. **1st of 26 teams** on the organisers' private test set of 20K real ID documents, in both tracks, and 2nd overall in manipulation localization.
- Introduced the **Membership Inference Test (MINT)**, a family of methods that lets an auditor with access to a model check whether given samples were in its training set. Activation-based MINT reaches up to 90% accuracy on 22M+ face images (IEEE Access 2025); Active MINT trains the model and its auditor jointly and stays above 80% from MobileNet to ViT (ICCV 2025); gMINT uses per-sample gradient updates and reaches near-perfect detection in most settings (CVPR-W 2025).
- Co-developed MINT for language models: 7 Transformer models and 2.5M sentences, AUC 0.85–0.99 (IEEE CAI 2026), and LoRA-fine-tuned LLMs, precision 0.77–0.92 across 4 models (preprint, 2026).
- Built **ai-mintest.org**, a public web tool that runs these audits on image and text models (AAAI 2025 workshop demo; IEEE COMPSAC 2026).
- Other first-author work: AttZoom, a drop-in spatial attention layer for CNNs (ICCV-W 2025); N-Sigma, a statistical framework for measuring bias in AI models (IEEE COMPSAC 2023).

ReTeX – Independent Research Project

Personal project

2026 – present

- Training a vision-language model that turns images of document pages into LaTeX source, treating document conversion as image-to-code generation. Paper in preparation.

EDUCATION

Universidad Autónoma de Madrid (UAM), Madrid, Spain

Ph.D. in Telecommunications and Computer Engineering

Jan 2022 – expected early 2027

Thesis: *Auditing Artificial Intelligence: Membership Inference and Methods for Transparent, Fair and Trustworthy AI*. Advisors: Julian Fierrez and Aythami Morales. Funded by an FPU fellowship (FPU21/05785), Spain's national 4-year predoctoral fellowship.

M.Sc. in Deep Learning for Audio and Video Signal Processing

2020 – 2021

Thesis: egocentric video localization and multi-user graph modelling for contact tracing with life-logging cameras.

B.Sc. in Telecommunications Engineering

2016 – 2020

Ranked 1st in the graduating class. 8 *Matrículas de Honor* (top distinction, capped at 5% of each course). UAM Excellence Scholarship; EPS Effort Award.

PUBLICATIONS

PDFs at danieldealcala.github.io/publications. *Equal contribution.

CONFERENCE AND JOURNAL PAPERS

- Active Membership Inference Test (aMINT): Enhancing Model Auditability with Multi-Task Learning. **D. DeAlcala**, A. Morales, J. Fierrez, G. Mancera, R. Tolosana, J. Ortega-Garcia. *ICCV 2025*.
- Is My Data in Your AI? Membership Inference Test (MINT) Applied to Face Biometrics. **D. DeAlcala**, A. Morales, J. Fierrez, G. Mancera, R. Tolosana, J. Ortega-Garcia. *IEEE Access*, 2025.
- DeepID Challenge of Detecting Synthetic Manipulations in ID Documents. P. Korshunov, . . . , **D. DeAlcala**, . . . (34 authors). *ICCV 2025*.
- Is My Text in Your AI Model? Gradient-based Membership Inference Test Applied to LLMs. G. Mancera, **D. DeAlcala**, J. Fierrez, R. Tolosana, A. Morales. *IEEE CAI 2026*.
- Is My Vision-Language Data in Your AI? Membership Inference Test (MINT) Demo 2. **D. DeAlcala***, G. Mancera*, J. Fierrez, A. Morales, R. Tolosana, R. Vera-Rodriguez. *IEEE COMPSAC 2026*.
- Measuring Bias in AI Models: An Statistical Approach Introducing N-Sigma. **D. DeAlcala**, I. Serna, A. Morales, J. Fierrez, J. Ortega-Garcia. *IEEE COMPSAC 2023*.

IN SUBMISSION AND PREPRINTS

- DocAI: Detecting and Localizing AI-Edited Content in Documents. **D. DeAlcala**, C.-Y. Ko, A. Morales, J. Fierrez, R. Tolosana, P.-Y. Chen. *Under review at ICLR 2027*.
- Auditing Training Data in Domain-adapted LLMs: LoRA-MINT. G. Mancera, **D. DeAlcala**, A. Morales, J. Fierrez, R. Tolosana, F. Jurado. *arXiv:2606.06946*, 2026.

WORKSHOP PAPERS

- Comparative Study of Domain-adapted VLMs for General Document Visual Question Answering. M. Lopez-Duran, E. Marrero, J. Fierrez, M. Robledo-Moreno, R. Vera-Rodriguez, **D. DeAlcala**, A. Morales, R. Tolosana, O. Delgado, A. Ortigosa, J. Ortega-Garcia. *ICDAR 2026 Workshops*.
- CrimeNER Demo: Named-Entity Recognition in the Crime Domain. M. Lopez-Duran, J. Fierrez, A. Morales, **D. DeAlcala**, G. Mancera, J. Irigoyen, R. Tolosana, O. Delgado, F. Jurado, A. Ortigosa. *ICDAR 2026 Workshops*.
- AttZoom: Attention Zoom for Better Visual Features. **D. DeAlcala**, A. Morales, J. Fierrez, R. Tolosana. *ICCV 2025 Workshops*.
- gMINT: Gradient-based Membership Inference Test Applied to Image Models. **D. DeAlcala**, A. Morales, J. Fierrez, G. Mancera, R. Tolosana. *CVPR 2025 Workshops*.
- MINT-Demo: Membership Inference Test Demonstrator. **D. DeAlcala**, A. Morales, J. Fierrez, G. Mancera, R. Tolosana, R. Vera-Rodriguez. *AAAI 2025 Workshop on AI Governance*.
- Membership Inference Test: Auditing Training Data in Object Classification Models. G. Mancera, **D. DeAlcala**, A. Morales, R. Tolosana, J. Fierrez. *AAAI 2025 Workshop on Deployable AI*.
- A Comprehensive Analysis of Factors Impacting Membership Inference. **D. DeAlcala**, G. Mancera, A. Morales, J. Fierrez, R. Tolosana, J. Ortega-Garcia. *CVPR 2024 Workshops*.
- BeCAPTCHA-Type: Biometric Keystroke Data Generation for Improved Bot Detection. **D. DeAlcala**, A. Morales, R. Tolosana, A. Acien, J. Fierrez, S. Hernandez, M. A. Ferrer, M. Diaz. *CVPR 2023 Workshops*.
- IFBiD: Inference-Free Bias Detection. I. Serna, **D. DeAlcala**, A. Morales, J. Fierrez, J. Ortega-Garcia. *AAAI 2022 Workshop on AI Safety (SafeAI)*.
- ALEBk: Feasibility Study of Attention Level Estimation via Blink Detection Applied to e-Learning. R. Daza, **D. DeAlcala**, A. Morales, R. Tolosana, R. Cobos, J. Fierrez. *AAAI 2022 Workshop on AI for Education*.

SERVICE AND TEACHING

- **Reviewer** for NeurIPS, ICML, ICLR, CVPR and ICCV (2023 – 2026); also AAAI workshops and IEEE T-BIOM.
- **Teaching assistant**, Multimedia Signal Processing, B.Sc. in Telecommunications Engineering, UAM (2023 – present).